Understanding the Film Formation Kinetics of Sequential Deposited Narrow-Bandgap Pb–Sn Hybrid Perovskite Films

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Developing efficient narrow bandgap Pb–Sn hybrid perovskite solar cells with high Sn-content is crucial for perovskite-based tandem devices. Film properties such as crystallinity, morphology, surface roughness, and homogeneity dictate photovoltaic performance. However, compared to Pb-based analogs, controlling the formation of Sn-containing perovskite films is much more challenging. A deeper understanding of the growth mechanisms in Pb–Sn hybrid perovskites is needed to improve power conversion efficiencies. Here, in situ optical spectroscopy is performed during sequential deposition of Pb–Sn hybrid perovskite films and combined with ex situ characterization techniques to reveal the temporal evolution of crystallization in Pb–Sn hybrid perovskite films. Using a two-step deposition method, homogeneous crystallization of mixed Pb–Sn perovskites can be achieved. Solar cells based on the narrow bandgap (1.23 eV) $FA_{0.66}MA_{0.34}Pb_{0.5}Sn_{0.5}I_3$ perovskite absorber exhibit the highest efficiency among mixed Pb–Sn perovskites and feature a relatively low dark carrier density compared to Sn-rich devices. By passivating defect sites on the perovskite surface, the device achieves a power conversion efficiency of 16.1%, which is the highest efficiency reported for sequential solution-processed narrow bandgap perovskite solar cells with 50% Sn-content.

1. Introduction

Organic–inorganic hybrid perovskites hold great promise as light-absorbers for low-cost and high-efficiency solar cells owing to their exceptional optoelectronic characteristics such as high absorption coefficient, long carrier diffusion length, and large defect tolerance.[1–3] Within a decade, the power conversion efficiency (PCE) of single-junction perovskite solar cells (PSCs) has increased from 3.8% to a record of 25.2%, on par with the state-of-the-art inorganic photovoltaic (PV) technologies.[4] Developing all-perovskite tandem solar cells with complementary bandgaps is considered as the next step to surpass the Shockley–Queisser efficiency limit for single-junction devices.[5–8] Solution processability of all components at low-temperature would enhance incentives to explore large-scale implementation of such multi-junction cells.[9]

Previous studies[10–12] of tandem solar cells have suggested that a combination of a wide-bandgap (1.7 to 1.9 eV) front cell and a narrow-bandgap (0.9 to 1.2 eV) rear cell is ideal for high efficiency multijunctions. Hybrid perovskites AMX_3, where A is a monovalent cation (formamidinium (FA^+), methylammonium (MA^+) or Cs^+), M is a divalent metal cation (Pb^{2+} or Sn^{2+}), and X is a halide anion (I^- or Br^-),[13] have a broadly tunable bandgap via compositional engineering.[14] The bandgap of Pb-based perovskites can be continuously tuned from 1.5 to 2.3 eV by substituting I with Br.[15,16] On the other hand, a nonlinear bandgap behavior[17] is observed when alloying Pb (1.5 eV) and Sn-based (1.3 eV) perovskites, providing a minimum bandgap of ≈1.2 eV at 50% to 75% Sn-content.[18–21] Such mixed Pb–Sn perovskites are considered the most promising narrow-bandgap perovskite materials for tandem devices.[21,22] However, compared to their Pb analogs, Sn-based perovskite precursors have a greater tendency to react and crystallize at room temperature,[23] which makes it difficult to grow compact, smooth, and homogeneous

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Pb–Sn hybrid perovskite films.[24,25] It has also been shown that readily oxidation of Sn^{2+} to Sn^{4+} can drastically increase trap density of the perovskite absorber and deteriorate performance of mixed Pb–Sn PSCs.[8,26] Strategies such as precursor engineering have since been exploited for high-quality Pb–Sn perovskite films.[12,27] Most recently, by adding guanidinium thiocyanate and metallic Sn powder to the precursor solution, high PCEs of 20.2% and 21.1%, respectively, have been demonstrated in mixed Pb–Sn low bandgap PSCs.[5,8]

Notably, most high-quality Pb–Sn perovskite films to date are prepared by the antisolvent method.[12,28] Despite being efficient in producing dense, crystalline perovskite films, the laborious antisolvent dripping procedure presents a potential risk of irreproducibility among different perovskite recipes and is considered to be less compatible with large-scale processes.[29,30] In comparison, a sequential deposition method provides better morphological control for Pb-based perovskites since the crystallization originates from an intermediate PbI_2 framework.[31,32] Furthermore, separate optimizations for each step are possible, which leads to a broader processing window with improved reproducibility.[33] However, only a handful of studies have used this method for Pb–Sn hybrid perovskite films,[26,34–38] and the highest PCE (16.26%)[26] is lower than achieved with the antisolvent method. Moreover, among sequentially deposited, high-performing Pb–Sn perovskites most films contain less than 30% Sn, leading to suboptimal bandgaps for tandem applications.[26,34,35] Understanding the growth mechanism of sequential solution-processed Pb–Sn hybrid perovskite films is critical for further optimizing high-performing Pb–Sn perovskite films with higher Sn-content.

In this study, we explore the compositional space of high-quality $FA_{0.66}MA_{0.34}Pb_{1-x}Sn_xI_3$ perovskite films prepared using a simple sequential deposition method. Through in situ optical absorption spectroscopy during spin coating, we reveal the impact of the Pb/Sn ratio on the film formation kinetics of hybrid perovskite films. Sn-containing precursors are found to accelerate the conversion to perovskite phase. Combining with ex situ characterization, we evidence the direct formation of large perovskite grains after coating the Sn-based precursor solutions. In comparison, only small perovskite nuclei are formed for the Pb-only precursor films. The sequential deposition approach enables the formation of homogeneous Pb–Sn hybrid perovskite phases despite the very different crystallization behavior of the Pb- and Sn-only perovskite precursors. We demonstrate compact and smooth perovskite films with Sn-contents up to 60%. After optimization, the narrow-bandgap (1.23 eV) $FA_{0.66}MA_{0.34}Pb_{0.5}Sn_{0.5}I_3$ perovskite provides a current density of 28.0 mA cm^{-2} and a PCE of 16.1% in a planar solar cell.

2. Results and Discussion

Figure 1a illustrates the fabrication of $FA_{0.66}MA_{0.34}Pb_{1-x}Sn_xI_3$ (x = 0, 0.25, 0.4, 0.5, 0.6, 0.75, 1) perovskite films via sequential deposition. First, a solution mixture of (1 − x) PbI_2 and x SnI_2 (denoted as $Pb_{1-x}Sn_xI_2$) in *N,N*-dimethylformamide (DMF) and dimethyl sulfoxide (DMSO) is spin coated to obtain the intermediate precursor film. The film is kept at room temperature for some time to evaporate excess solvents. The film is in an amorphous state (Figure S1, Supporting Information) that remains for hours due to the strong coordination between SnI_2 and DMSO.[23] In a second step, an isopropanol solution of FAI/MAI (66:34 in molar ratio) is spin coated on top, followed by thermal annealing to promote the crystallization of the perovskite film.

To gain insight into transformation dynamics of the $Pb_{1-x}Sn_xI_2$ precursor, we performed in situ optical absorption measurements during spin coating of the FAI/MAI solution. Following a previously described protocol,[39] the temporal evolution of the absorption spectrum was recorded by collecting the light that was reflected from rotating substrates with a white reflector at the back (Figure S2, Supporting Information). Figure 1b exhibits the absorption spectra of a typical $FA_{0.66}MA_{0.34}Pb_{0.5}Sn_{0.5}I_3$ (x = 0.5) precursor film at different times during spin coating. Directly after dripping the FAI/MAI solution (t = 0 s), the absorbance increases abruptly over

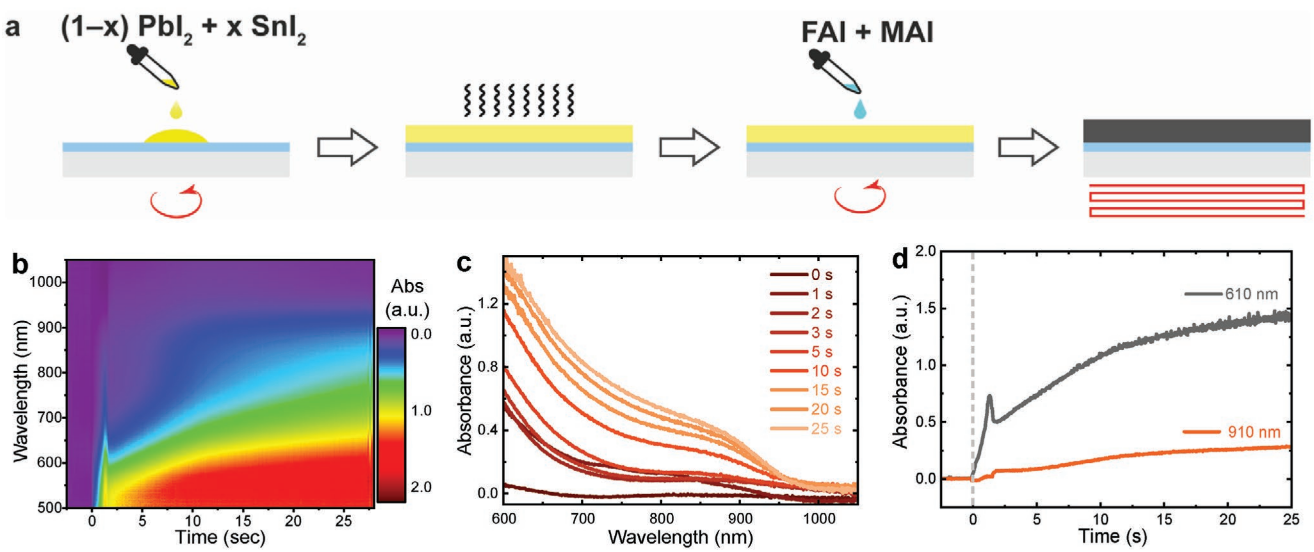

Figure 1. a) Schematic fabrication process of $FA_{0.66}MA_{0.34}Pb_{1-x}Sn_xI_3$ perovskite films using a sequential deposition method. b) Time-resolved absorption intensity maps during spin coating of FAI/MAI solution on the $Pb_{0.5}Sn_{0.5}I_2$ precursor film. c) Corresponding absorption spectra. d) Time-evolution of absorbance at 610 and 910 nm during the spin coating process.

a broad wavelength region, giving rise to the characteristic narrow-bandgap perovskite spectrum at the earliest stages of the deposition (Figure 1c). The absorption intensity then slightly decreases when the material is radially ejected from the substrate during the flow phase ($t < 3$ s).[39] Subsequently, during the drying phase ($t < 30$ s), the optical absorption increases owing to the formation of perovskite phase. This implies that the transition from the amorphous $Pb_{0.5}Sn_{0.5}I_2$ precursor to the crystalline perovskite is initiated by casting the FAI/MAI solution. Also, the simultaneous increase of absorbance in both the visible (610 nm) and near-infrared (910 nm) regions suggests that there is no significant preferential formation of segregated Pb- and Sn-rich perovskite phases (Figure 1d).

When comparing in situ absorption spectra of different mixed Pb–Sn compositions (Figure S3, Supporting Information), we found that the conversion to perovskite phase is accelerated at higher Sn contents. For the Sn-only precursor film ($x = 1$), a sharp absorption onset at around 900 nm is observed immediately after dripping the FAI/MAI solution. Afterward, the absorbance is initially lowered due to the layer thinning and then remains constant. This implies that the formation of a Sn-only perovskite phase is completed, directly after casting the FAI/MAI solution. On the contrary, the transformation to perovskite phase is considerably slow for the Pb-only composition ($x = 0$), as evidenced by the weak absorption throughout the deposition process. Ex situ SEM and XRD measurements for as-cast (nonannealed) films (Figures S4 and S5, Supporting Information) confirm that only small-sized nanocrystals were formed in the Pb-only precursor film ($x = 0$), which correlates with slow crystal growth of Pb-based nuclei before the thermal annealing. In comparison, the Sn-only precursor film exhibits a much higher crystallinity, accompanied by the formation of micrometer-sized grains on the surface. The results confirm the higher affinity of the organic components FAI/MAI toward SnI_2, which facilitates crystal growth of Sn-based perovskite at room temperature.

The crystallization behavior of Pb–Sn based precursors has a significant impact on the quality of the final perovskite films. **Figure 2**a–c and Figure S6 (Supporting Information) display top-view SEM images of the annealed perovskite samples with different Pb/Sn ratios. Compared to nonannealed precursor films (Figure S4, Supporting Information), thermal annealing causes coarsening of the perovskite crystals. It is found that all annealed films display a compact and pinhole-free surface morphology, with the average grain size starting at 240 ± 80 nm for the Pb-only perovskite film ($x = 0$), 530 ± 140 nm for $Pb_{0.5}Sn_{0.5}$ ($x = 0.5$) and 1.3 ± 0.3 µm for Sn-only ($x = 1$) perovskite films (Figure S7, Supporting Information). Meanwhile, atomic force microscopy (AFM) (Figure S8, Supporting Information) shows that the root-mean-square (RMS) surface roughness of the Sn-rich perovskite films ($x > 0.5$) is considerably higher than that of the Pb-rich films ($x \leq 0.5$), in line with their propensity to form large crystallites at room temperature.

XRD patterns confirm the formation of single-phase Pb–Sn hybrid perovskite crystallites in sequentially deposited $FA_{0.66}MA_{0.34}Pb_{1-x}Sn_xI_3$ systems (Figure 2g). Following crystallographic data reported by Kanatzidis and co-workers,[40] we indexed all Pb–Sn hybrid perovskites to the cubic space group $Pm\overline{3}m$ and determined the corresponding lattice parameters (Figure S9 and Table S1, Supporting Information). When replacing Pb^{2+} with smaller Sn^{2+} (x from 0 to 1),[41] the (100)/(200) diffraction peaks gradually shift toward higher angles, corresponding to a monotonic decrease of the cubic lattice constant from 6.34 to 6.27 Å (Figure S10, Supporting Information). Interestingly, a similar trend can also be found in the non-annealed precursor films, of which the lattice constant decreases linearly with the increasing Sn content (Figure S10, Supporting Information). Our observation indicates that despite the very different crystallization behavior of the Pb- and Sn-based perovskite precursors, uniform Pb–Sn hybrid perovskite phases can be formed before thermal annealing. This can be attributed to the amorphous intermediate film used in the sequential deposition process. During the FAI/MAI solution dripping, the intermixed PbI_2 (1–x) and SnI_2 (x) phases enable a simultaneous and yet homogeneous transformation to the Pb–Sn hybrid perovskite crystallites. In optical absorption spectra, the characteristic bandgap bowing[17] is seen for both non-annealed and annealed Pb–Sn perovskite (Figure S11, Supporting Information). This result also indicates that homogeneous crystallization has occurred. The narrowest bandgap of ≈1.23 eV is obtained between 50% and 75% Sn contents, whereas the pure Pb- and Sn-compounds show bandgaps of 1.55 and 1.44 eV, respectively.

It has been reported that the perovskite crystal orientation depends on the microstructure of the as-deposited precursor film.[41,42] Figure 2d–f and Figure S12 (Supporting Information) show the grazing-incidence wide-angle X-ray scattering (GIWAXS) patterns of different Pb–Sn hybrid perovskite films. All Sn-containing perovskite films ($x \geq 0.25$) exhibit isotropic diffraction rings for the various lattice spacings, suggesting no preferred orientation of the Pb–Sn hybrid perovskite crystals.[43] This is attributed to the predominant growth of randomly oriented crystals in the precursor film (Figure S4, Supporting Information), which suppresses re-orientation during thermal annealing. In Figure 2e, the scattering feature at $q = 0.91$ Å^{-1} ($2\theta = 12.7°$) corresponds to the PbI_2 phase, which originates from degradation during sample transfer in ambient air to the X-ray system. In comparison, a varying reflection intensity and more distinct Bragg spots along the diffraction ring of the (100) plane ($q = 1$ Å^{-1}) are observed for the Pb-only perovskite film ($x = 0$), indicating a slightly higher degree of texture (Figure S13, Supporting Information).[44] We anticipate that during the thermal annealing, the pure Pb-based nuclei have more freedom to grow along the preferred directions.

Improving the homogeneity of the perovskite film is of critical importance to the electrical properties such as defect density and carrier lifetime.[24,45] To investigate the phase distribution of Pb–Sn hybrid perovskite films using sequential deposition method, we carried out a detailed microstructure analysis. First, surface X-ray photoelectron spectroscopy (XPS) reveals that the Sn/Pb atomic ratio matches the composition used in precursor solutions. A monotonic increase in the Sn atomic concentration can be found as x increases (**Figure 3**a). We note that even when the XPS samples were prepared and transferred in a N_2 atmosphere, a small trace of oxygen (O 1s) can be found in all Pb–Sn hybrid perovskite films (Figure S14, Supporting Information). Deconvolution of the Sn 3d peaks suggests that the formation of Sn^{4+} species is largely uncontrolled (5–10 at%) among different Pb/Sn ratios, which confirms the ease of oxidation of Sn^{2+} in the Sn-containing perovskite films.[8]

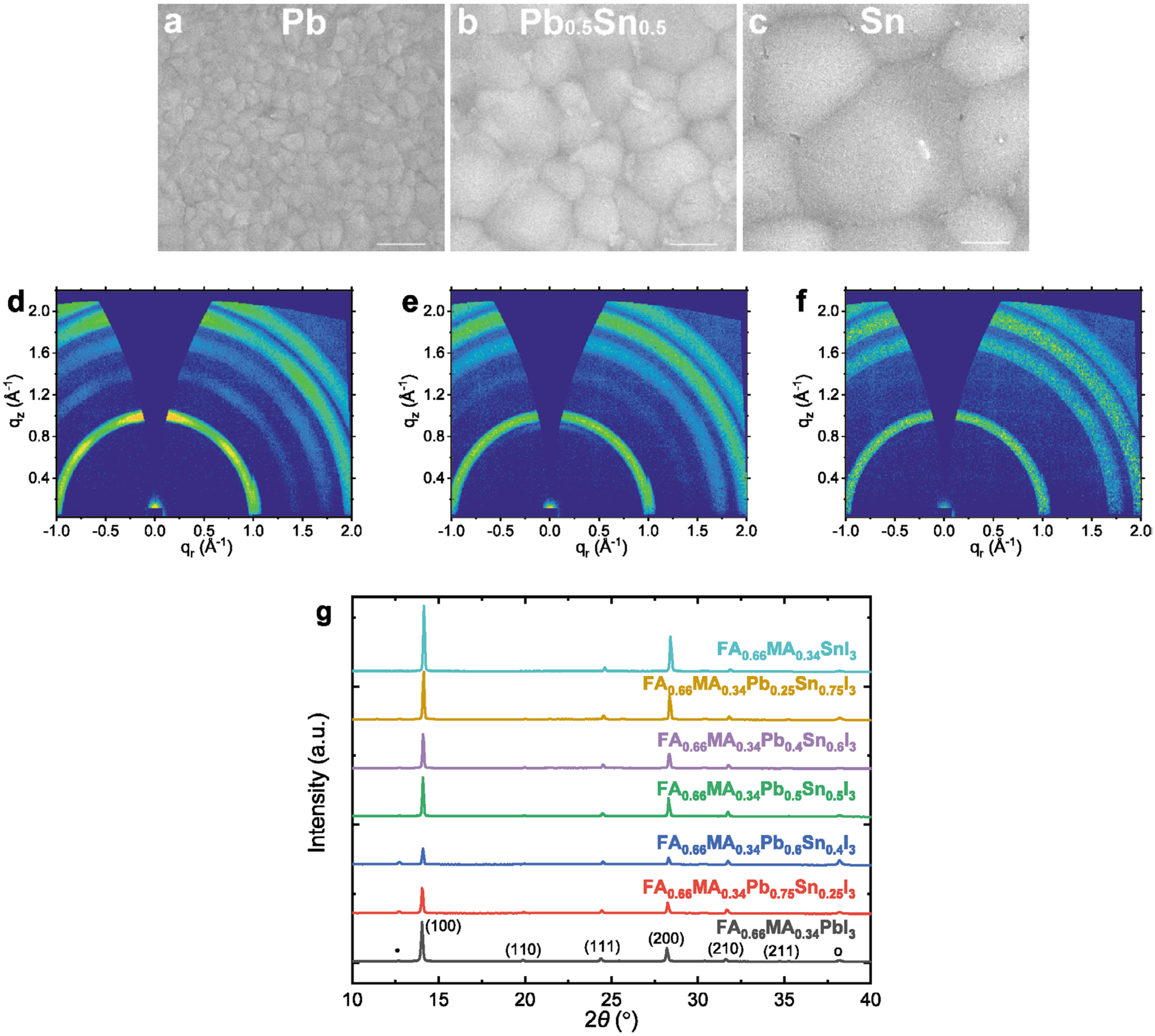

Figure 2. a–c) Top-view SEM images and d–f) 2D GIWAX patterns of annealed $FA_{0.66}MA_{0.34}Pb_{1-x}Sn_xI_3$ perovskite films, where $x = 0$ (a,d), $x = 0.5$ (b,e), and $x = 1$ (c,f). g) XRD patterns of annealed $FA_{0.66}MA_{0.34}Pb_{1-x}Sn_xI_3$ perovskite films with different Pb–Sn ratios, which are indexed to the cubic space group $Pm\bar{3}m$. Scale bars in SEM images are 500 nm.

Furthermore, cross-sectional scanning transmission electron microscopy (STEM) and energy-dispersive X-ray spectroscopy (EDS) analysis were performed on a PSC based on $FA_{0.66}MA_{0.34}Pb_{0.5}Sn_{0.5}I_3$ absorber (the device performance will be discussed later). The high-angle annular dark-field (HAADF) STEM image (Figure 3b) shows well-defined interfaces between different layers over the cross-section region and the perovskite film exhibits large grains that span the entire thickness of the active layer (≈400 nm). The conformal and homogeneous hole- and electron-transport layers of poly(3,4-ethylenedioxythiophene) polystyrene sulfonate) (PEDOT:PSS) and C_{60}/bathocuproine (BCP) prevent the perovskite film from making direct contact to the indium tin oxide (ITO) and Ag electrodes, respectively. In line with the TEM image, EDS elemental mappings acquired in the same region also confirm the integrity of the device stack. The Ag signal is confined to the top electrode region. The intense C signal located at both perovskite interfaces correspond to the C_{60}/BCP (top) and PEDOT:PSS (bottom) layers, respectively. The relatively homogeneous distribution of Sn, Pb, I, and C suggests the high quality of the Pb–Sn perovskite film prepared by sequential deposition method. We note that the formation of small PbI_2-rich domains (displayed as brighter regions highlighted along circles and defined lines in Figure 3c) is mainly due to sample degradation induced by prolonged air exposure.[46] The result is also confirmed by an EDS line-scan along the same cross-sectional region (Figure S15, Supporting Information), where both the Pb and I signals display a peak near the center of the perovskite layer.

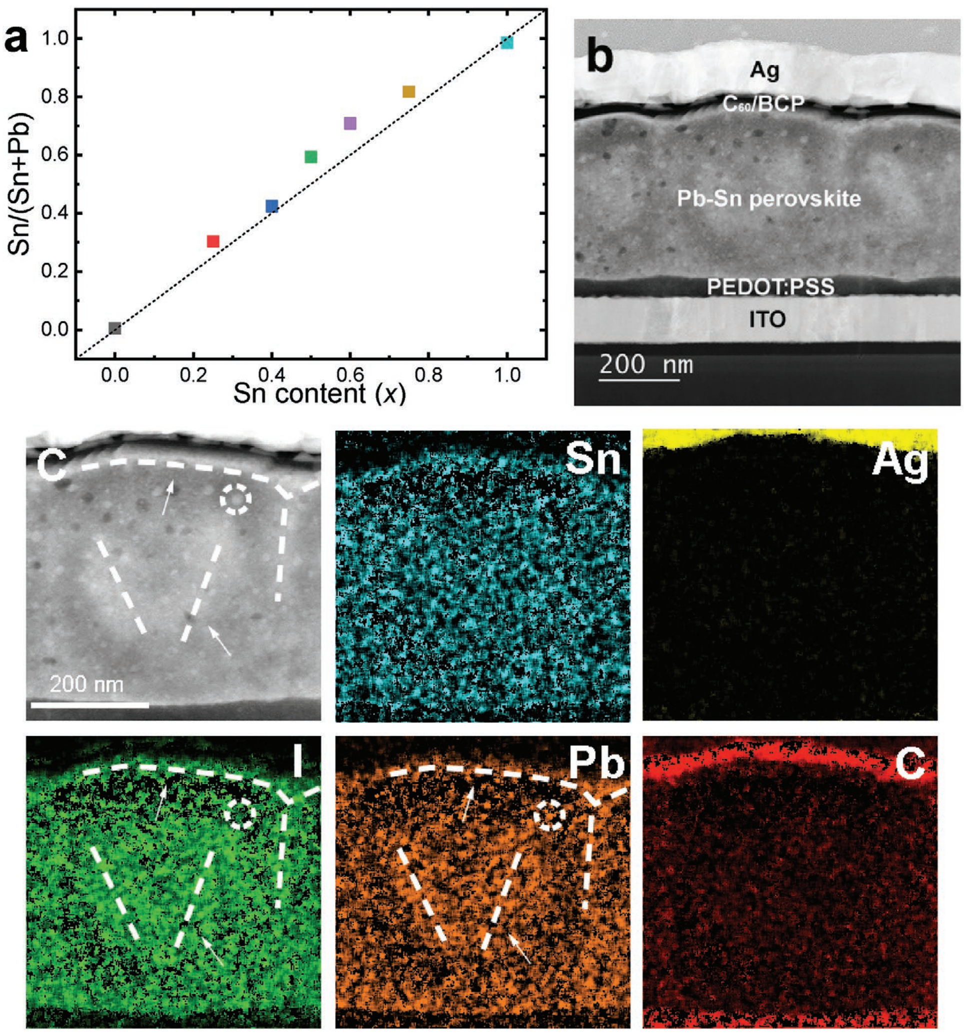

Figure 3. a) The atomic ratio Sn/(Sn + Pb) measured by XPS versus the nominal composition used in the precursor solutions. The atomic ratio varies from 0 to 1. b) HAADF cross-sectional STEM image. c) Corresponding EDS elemental maps of a PSC based on the ITO/PEDOT:PSS/$FA_{0.66}MA_{0.34}Pb_{0.5}Sn_{0.5}I_3/C_{60}$/BCP/Ag structure. To reduce the background X-ray artifacts, quantitative EDS mapping was performed by binning 5 × 5 pixels for each pixel to be quantified (see details in the Experimental Section, Supporting Information). Circles and dashed lines drawn in TEM image and Pb/I elemental maps highlight the formation of PbI_2-rich domains due to sample degradation.

To evaluate the PV performance of sequential processed Pb–Sn hybrid perovskites, we fabricated planar p-i-n PSCs using an ITO/hole transport layer (HTL)/$FA_{0.66}MA_{0.34}Pb_{1-x}Sn_xI_3/C_{60}$/BCP/Ag device structure. For the Pb-only device, poly(triaryl amine) (PTAA) was used as the HTL, whereas PEDOT:PSS was used for Sn-containing solar cells ($x \geq 0.25$). **Figure 4**a displays the stabilized current density–voltage (J–V) curves of PSCs with different Sn-content and the corresponding photovoltaic parameters are summarized in Table S2 (Supporting Information). The external quantum efficiency (EQE) spectra were measured with ≈1 sun equivalent bias illumination to provide more accurate data for the short-circuit current density (J_{sc}) and PCE (Figure 4b). It is found that the pure Pb-based ($x = 0$) device exhibits a PCE of 17.6%, with a J_{sc} of 21.8 mA cm^{-2}, an open-circuit voltage (V_{oc}) of 1.02 V and a fill factor (FF) of 0.79. As the Sn-content reaches 50% ($x = 0.5$), the J_{sc} gradually increases to 26.9 mA cm^{-2}, whereas V_{oc} and FF drop to 0.74 V and 0.71, respectively, resulting in the highest PCE (of 14.1%) among the Pb–Sn hybrid PSCs. The changes in J_{sc} and V_{oc} can be attributed to the narrowed bandgap after incorporation of Sn as seen in the EQE onset that redshifts from 840 nm ($x = 0$) to 1040 nm ($x = 0.5$). The decrease in FF can result from the uncontrolled p-doping induced by Sn^{2+} oxidation to Sn^{4+} (Figure S14, Supporting Information), which significantly reduces the charge carrier diffusion length of Sn-based perovskite films.[8,45] While the PSC with 60% Sn still shows a PCE of 13.7%, the performance quickly deteriorates upon further increasing Sn-content ($x \geq 0.75$). We expect that the rough Sn-rich perovskite films (Figure S8, Supporting Information) introduce shunting pathways and thereby more recombination losses in the solar cells.

To estimate the background carrier density of Pb–Sn hybrid PSCs, we performed capacitance–voltage (C–V) measurements under dark conditions.[47,48] According to Mott–Schottky plots (Figure S16, Supporting Information), pure Pb-based ($x = 0$) PSC exhibits a carrier density of 1.91×10^{15} cm^{-3}, which slightly increases to 3.95×10^{15} cm^{-3} for 50% Sn-based PSC ($x = 0.5$),

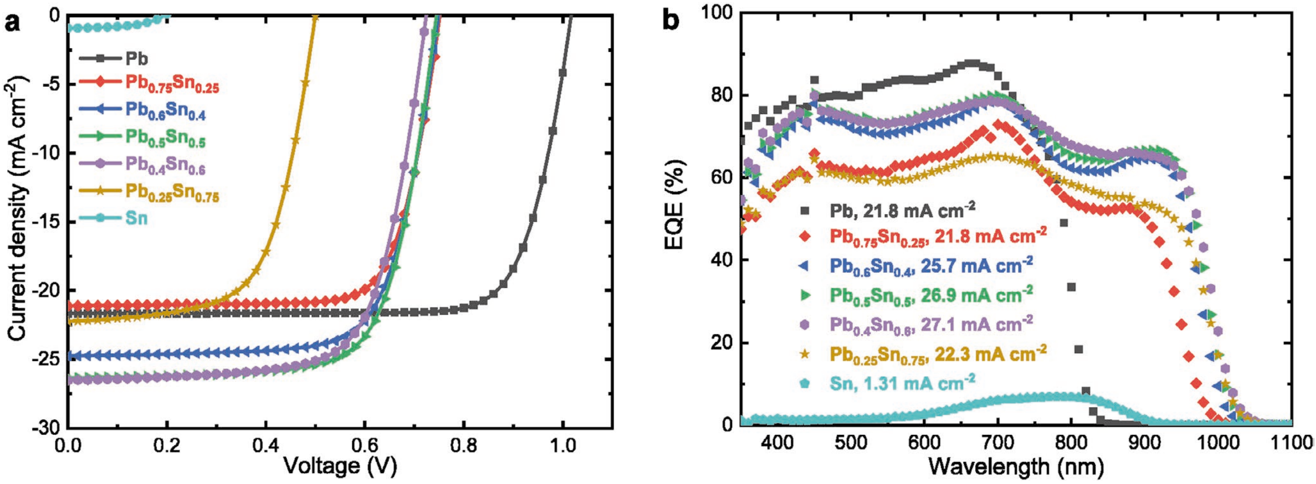

Figure 4. a) Stabilized *J–V* curves of the best-performing PSCs based on ITO/PTAA or PEDOT:PSS/$FA_{0.66}MA_{0.34}Pb_{1-x}Sn_xI_3/C_{60}$/BCP/Ag device structure, where x = 0, 0.25, 0.4, 0.5, 0.6, 0.75, and 1. b) Corresponding EQE spectra. The inset lists the current densities obtained by integrating the EQE with the AM1.5G spectrum.

and quickly ramps up to 1.06×10^{16} cm^{-3} for pure Sn-based PSC ($x = 1$). The high background carrier density of Sn-rich PSCs is consistent with the high leakage current density extracted from dark *J–V* curves (Figure S17, Supporting Information). Based on the discussion above, we conclude that the increased dark carrier density can be mainly attributed to shunts in Sn-rich PSCs, which deteriorates device performance.

To further improve the performance of $Pb_{0.5}Sn_{0.5}$-based PSCs, we introduced a post-deposition treatment by spin coating an isopropanol-based ammonium thiocyanate (NH_4SCN) solution onto the perovskite film. Through optimization of processing conditions (Figure S18, Supporting Information), the best performance is obtained when using a 1 mg mL^{-1} of NH_4SCN solution without additional thermal annealing. Compared to the pristine $Pb_{0.5}Sn_{0.5}$-based PSCs, solar cells with NH_4SCN treatment exhibits a much higher PCE of 15.9%, with a J_{sc} of 28.0 mA cm^{-2}, a V_{oc} of 0.78 V, and an FF of 0.73 (**Figure 5**a,b and Table S3, Supporting Information). Meanwhile, almost identical *J–V* curves were measured in reverse, forward, and stabilized scans, indicating negligible hysteresis (Figure 5e). NH_4SCN-treated PSC shows a stabilized PCE of 16.1% after 300 s of steady-state power output tracking (measured at the maximum power point V_{MPP} = 0.62 V, Figure 5f), which is the highest PCE reported for sequentially solution-processed narrow bandgap PSCs with 50% Sn content. Device statistics (Figure 5c,d and Figure S19, Supporting Information) suggest that the PCE enhancement is mainly due to an increase in V_{oc} from 0.70 ± 0.02 V for the control device to 0.76 ± 0.02 V for the NH_4SCN-treated device. From the light-intensity dependent V_{oc} measurements (Figure S20, Supporting Information), it is found that the NH_4SCH-treated device exhibits consistently higher V_{oc} at various illumination intensities. The Mott–Schottky analysis shows an increase in built-in voltage from 0.51 to 0.59 V when the perovskite layer was treated by NH_4SCN solution (Figure S21, Supporting Information), which correlates to a more efficient charge transfer process at the perovskite/ETL interface.[49,50]

We note that the improved device performance is not caused by changes in morphology and crystallinity, as both films display similar surface morphology and XRD patterns (Figure S22, Supporting Information). To study the effect of NH_4SCN treatment on surface composition of $Pb_{0.5}Sn_{0.5}$-based perovskite films, we performed XPS measurements. Compared to pristine perovskite films, the NH_4SCN-treated perovskite exhibits negligible changes in the spectral shape, and no signal from the S 2p peak can be detected (Figure S23, Supporting Information). This can be attributed to the use of a dilute NH_4SCN solution and the volatile nature of NH_4SCN. On the other hand, it is noted that the peak intensities of both I 3d and N 1s core levels are slightly increased, together with a shift of 91 meV toward higher peak binding energies. The small, positive shift in I 3d and N 1s binding energies can possibly indicate hydrogen bond formation between NH_4^+ and $[PbI_6]^{4-}$/$[SnI_6]^{4-}$ moieties.[51,52] The NH_4SCN treatment also results in an increase of the I/(Pb + Sn) atomic ratio from 1.97:1 for the pristine perovskite to 2.30:1 for the NH_4SCN-treated perovskite films. The NH_4SCN-treated perovskite film exhibits a higher photoluminescence intensity than the pristine perovskite (Figure S24, Supporting Information), indicating a reduced nonradiative recombination process. Although the exact reason why NH_4SCN treatment improves the performance is not fully clear, we expect that the number of defect sites at the perovskite surface is reduced after NH_4SCN treatment, reducing nonradiative recombination losses at the surface and thereby providing a higher V_{oc} in the Pb–Sn hybrid PSCs.

We proceeded to monitor the shelf lifetime of a laminated narrow bandgap PSC under ambient conditions (25 °C, 20–30 RH%). As shown in Figure S25 (Supporting Information), the device exhibited almost negligible decrease in the PCE (≈14.9%) over the tracking period, suggesting a good stability of such narrow bandgap perovskite material when air ingression is significantly reduced. We then fabricated a 4-terminal all-perovskite tandem solar cell by mechanically stacking a narrow bandgap back cell (1.23 eV) with a semitransparent front cell based on a 1.57 eV $FA_{0.66}MA_{0.34}PbI_{2.85}Br_{0.15}$ perovskite (Figure S26, Supporting Information). Here, the front cell utilizes a device structure of glass/ITO/PTAA/Perovskite/PCBM/Al-doped ZnO/ITO,

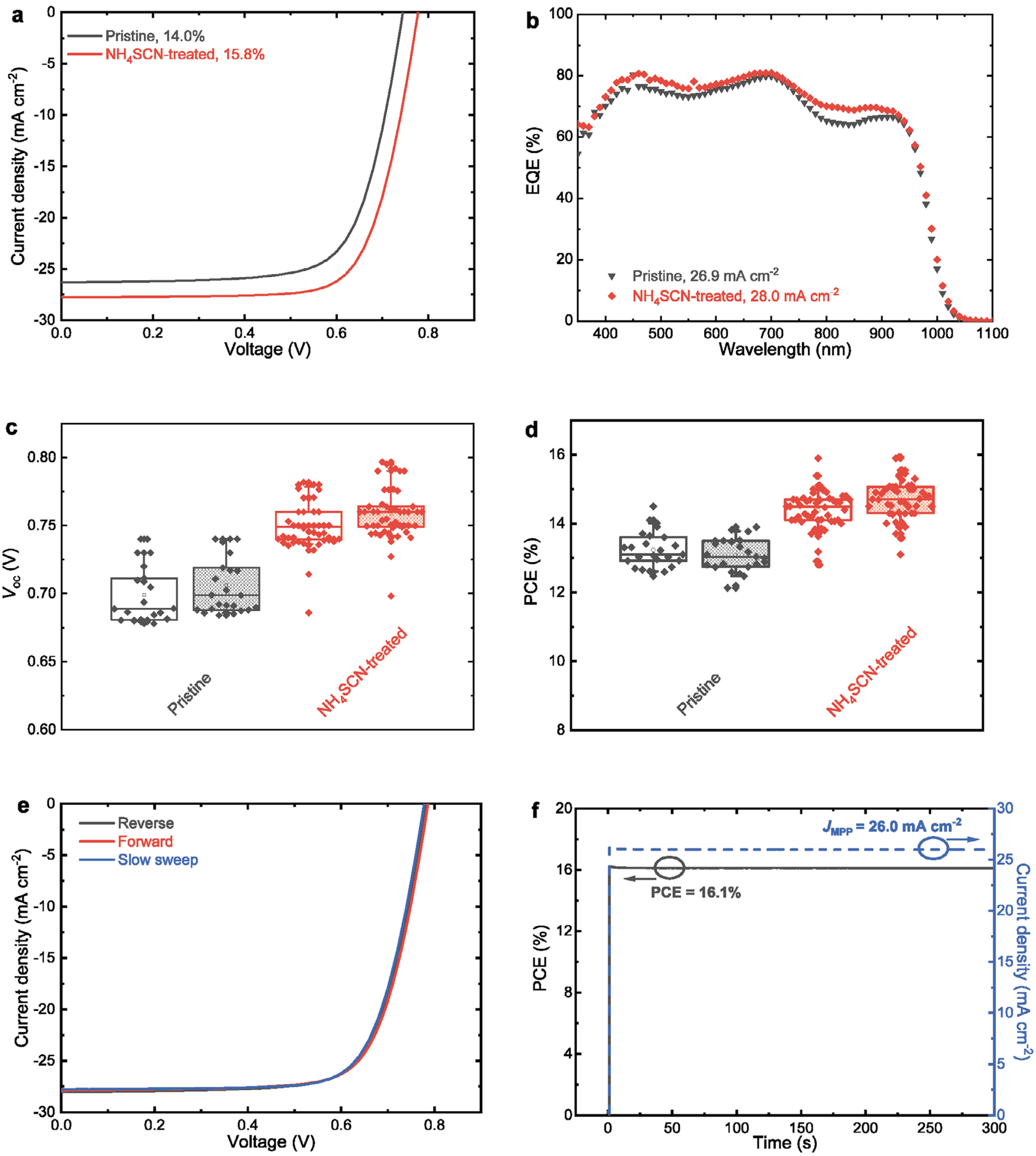

Figure 5. a) Stabilized J–V curves, b) EQE spectra with current densities obtained by integrating with the AM1.5G spectrum, and c,d) statistical distribution of V_{oc} and PCE of the PSCs based on $FA_{0.66}MA_{0.34}Pb_{0.5}Sn_{0.5}I_3$ without (pristine) and with NH_4SCN (1 mg mL^{-1}) post-treatment. The J–V measurements were fast sweeps in reverse (left) and forward (right) directions. e) J–V curves for the device with NH_4SCN post-treatment. f) Steady-state power/photocurrent output tracking for the device with NH_4SCN post-treatment.

where MgF_2 was deposited both on the glass side and the top ITO side to minimize the optical loss in the back cell. The semitransparent front cell shows a PCE of 16.0%, with a J_{sc} of 19.8 mA cm^{-2}, V_{oc} of 1.02 V, and FF of 0.79 (Table S4, Supporting Information). Meanwhile, the filtered narrow bandgap cell exhibits a PCE of 5.1%, with a J_{sc} of 9.3 mA cm^{-2}, V_{oc} of 0.72 V, and FF of 0.76. Compared to the stand-alone narrow bandgap cell, the EQE of the filtered cell shows a small decrease between 800 and 1000 nm, confirming a high transparency of the front cell in the near infrared. As a result, we achieved a high PCE of 21.1% in the 4-terminal tandem cell.

3. Conclusions

In summary, we have investigated the film formation and performance of Pb–Sn hybrid perovskites ($FA_{0.66}MA_{0.34}Pb_{1-x}Sn_xI_3$, $0 \leq x \leq 1$) deposited via a simple sequential deposition method. In situ absorption measurements, combined with ex situ characterizations, reveal the dynamic transformation of different Pb–Sn precursors and its impact on perovskite films. While the crystallization to pure-Sn perovskite phase is completed during casting the solution, the formation of the pure-Pb based perovskite is considerably slow without thermal annealing. We show that despite the different crystallization behavior of Pb- and Sn-based perovskite precursors, the use of an amorphous PbI_2/SnI_2 precursor film in the sequential deposition ensures a uniform transformation to the Pb–Sn hybrid perovskite phase. Owing to the high affinity between the SnI_2 and organic halides, the crystal growth at room temperature is accelerated with increasing Sn contents, in line with the enlarged grain sizes from ≈200 nm to over 1 µm for the pure-Pb and pure-Sn based perovskites, respectively. However, the fast crystal growth of the Sn-rich perovskite films ($x > 0.6$) also results in a higher surface roughness, which introduces large shunts and is hence detrimental to device performance. Meanwhile, we demonstrate smooth and homogeneous narrow bandgap (1.23 eV) perovskite films with 50% Sn content. Additional treatment with NH_4SCN is found to efficiently passivate defect sites on the perovskite surface, which reduces nonradiative recombination and improves V_{oc}. The best performing device based on a 460 nm thick $FA_{0.66}MA_{0.34}Pb_{0.5}Sn_{0.5}I_3$ exhibits a PCE of 16.1%, with a good stability under the ambient condition. Using this narrow bandgap cell, we demonstrated a 21.1% 4-terminal all-perovskite tandem solar cell.

Controlling the morphology of perovskite films is of critical importance for large-scale fabrication. Compared to its pure-Pb counterpart, the fast-crystallizing Sn-based perovskite imposes a greater processing challenge, for which the sequential deposition route with a broad processing window can be beneficial. Our study provides new insights into the film formation kinetics of the Pb–Sn hybrid perovskite films and highlights the potential of exploring high-efficiency narrow bandgap Pb–Sn PSCs using the sequential deposition method.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

crystallization, film formation, metal halide perovskites, narrow bandgap, solar cells

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